SUN TUNNEL

Tubular Daylighting Device (TDD)

Operating principle and detailed economic effectiveness



A Sun Tunnel can eliminate substantial daytime electric-light use in deep interior zones, but the financial result is usually controlled by three things: the watts actually switched or dimmed, occupied daylight hours, and the total installed roof-and-controls cost.

Scope: VELUX Commercial Sun Tunnel TCC/TGC, 14-inch and 22-inch rigid reflective-tube systems, plus general TDD economics. This is a planning guide, not a construction specification or quotation.

1. Executive summary

Technical name	The current technical term is Tubular Daylighting Device (TDD). Common market names include tubular skylight, light tube, sun pipe, and Sun Tunnel. NFRC 203 specifically addresses visible transmittance of tubular daylighting devices. [10]

What it does. A roof-mounted dome collects direct and diffuse sunlight, a highly reflective metal tube carries the light through the roof/ceiling cavity, and an interior diffuser spreads it into the occupied space. The passive optical path requires no electricity. Optional daylight controls use a photosensor and dimming/switching interface to reduce electric lighting.

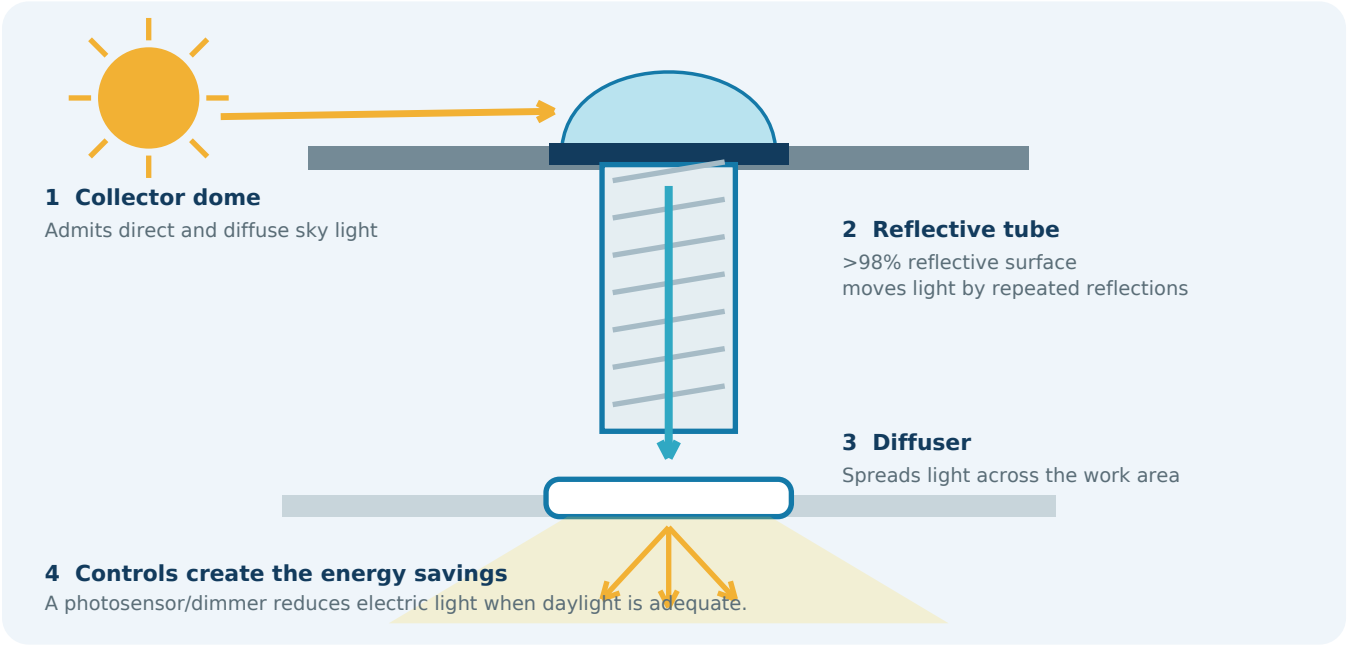
VELUX commercial configuration. Current VELUX information lists 14-inch and 22-inch systems. Product documents describe silver-coated aluminum rigid tubing with greater than 98% total reflectivity; the 22-inch SunCurve dome is intended to capture and redirect low-angle sunlight. TCC is curb-mounted/counter-flashed; TGC is self-flashed. Diffusers are available for suspended tile, hard, and open ceilings. [1]-[5]

Question	Practical answer
Can it reduce daytime lighting?	Yes - when daylight is sufficient at the work plane and electric lights are actually switched or dimmed. Without controls or disciplined manual switching, measured energy savings can be close to zero.
Is it always economical?	No. Pure energy payback can be long in LED buildings with low electricity prices. It becomes attractive when one device displaces a large controlled wattage for many occupied daylight hours, installation is simple, or non-energy benefits are valued.
What is the main design risk?	Overestimating delivered illuminance. Long tubes, multiple elbows, dirty domes, poor spacing, dark rooms, obstructions, and cloudy climate reduce useful daylight.
Best first action	Model the exact roof-to-work-plane geometry with VELUX IES files/Daylight Estimator, then connect each daylighted zone to automatic continuous dimming or multilevel switching.

Economic screen	For a $2,500 installed system evaluated over 20 years at 5% discount and 2% annual savings growth, annual first-year cash savings must be about $170 to reach break-even net present value. At the 2025 U.S. commercial electricity average of 13.41 cents/kWh, that usually requires a sizeable controlled lighting zone, demand savings, maintenance savings, or a lower installed cost. [7]

2. Operating principle

The base device is passive; the savings mechanism is active lighting control.



Stage	Physical action	Design consequence
1. Capture	The exterior dome receives beam sunlight and diffuse skylight. The 22-inch SunCurve form redirects rays that would otherwise pass through at low solar angles.	Avoid roof shadows from parapets, equipment, vegetation, and adjacent structures.
2. Transport	Light travels through a rigid, silver-coated aluminum tube by repeated specular reflections.	Use the shortest, straightest route. Every reflection, joint, elbow, and surface defect causes loss.
3. Distribute	The ceiling diffuser converts the directional tube output into a broad luminous pattern.	Choose diffuser and spacing for the task plane, ceiling type, glare control, and uniformity.
4. Control	A photosensor compares available daylight with a setpoint and dims or switches the electric-light zone.	Commission sensor location, setpoint, deadband, fade rate, and minimum light level; verify that occupants do not override it.

Why tube geometry matters: an idealized optical approximation is $T_{tube} \sim= rho^{N}$, where rho is wall reflectance and N is the average number of reflections. At rho = 0.98, 10 reflections retain about 82% of light at the tube wall, while 20 retain about 67%, before dome, joint, and diffuser losses. This is not a certified system rating; it explains why a short straight tube materially outperforms a long route with several elbows.

Important distinction

The tube delivers daylight; it does not by itself save electricity. Electricity is saved only when the corresponding electric-light load is reduced during occupied hours.

3. Product architecture and design variables

Item	VELUX commercial information	Implementation note
System types	TCC curb-mounted/counter-flashed; TGC self-flashed.	Coordinate curb, roof membrane, drainage, and roof warranty with the roofing manufacturer.
Nominal diameters	14 inches and 22 inches.	Do not select by diameter alone; size by simulated work-plane illuminance and annual daylight availability.
Tube	Rigid silver-coated aluminum, greater than 98% total reflectivity.	Keep protective film on during handling; seal joints; avoid dents, fingerprints, and gaps.
Dome	14-inch low profile; 22-inch high-profile SunCurve options; acrylic or impact-resistant polycarbonate variants.	Choose glazing/impact option for windborne-debris, hail, and local approval requirements.
Diffusers	TTC suspended tile, THC hard/drywall, TOC open ceiling; prismatic, frosted, or Fresnel options in product documents.	Check plenum, fire, suspension, glare, and cleaning requirements.
Tube length	Downloaded product sheets list up to 30 ft for TCC and 20 ft for TGC; the current sizing chart says lengths above 20 ft are not recommended.	Follow the current model-specific submittal and sizing chart, not a generic maximum.
Accessories	Daylight controller/dimmer, energy kit, vapor barrier, elbows, extensions, turret, security bars, fire band, suspension kit.	Include controls and envelope accessories in both design and cost.

Performance should be checked in a climate-based daylight model. VELUX provides a Daylight Estimator that uses product IES photometric files, space geometry, ceiling height, work-plane height, room reflectances, product count, and desired illuminance. The manufacturer suggests default reflectances of 80% ceiling, 50% walls, and 20% floor when project values are unknown. [3]

Performance and value drivers

High output / better economics	Low output / weaker economics
Short, vertical tube; few/no elbows	Long tube; several elbows or offsets
Unshaded roof exposure; clear sky access	Parapet/equipment shading; dense urban obstruction
High daytime occupancy and high controlled wattage	Mostly night use or already-low LED wattage
Automatic continuous dimming or reliable switching	No controls; occupants leave lights on
High electricity/demand rates	Low energy rates and no demand charge benefit
New construction or reroof project	Difficult retrofit, structural conflicts, expensive roof access

Recommended target

DOE guidance describes a good daylighting solution as providing at least 300 lux for at least 50% of operating hours while controlling glare and excessive heat. Treat that as a concept-screening target; the project illuminance criterion must match the actual task and code. [6]

4. How energy and cost savings are created

The annual savings calculation must start with the electric-light load that the TDD can actually control - not with dome area or a marketing lumen claim.

Annual lighting electricity saved

$kWh_{light} = P_{controlled}$ (kW) x $H_{occupied\text{-}daylight}$ (h/yr) x $F_{displacement}$

Net site electricity ~= kWh_{light} + cooling benefit - heating penalty. Annual cash savings = energy value + demand value + avoided maintenance.

Term	How to estimate it	Common error
Controlled power, P	Sum the input watts of fixtures in the daylight zone, multiplied by the actual dimming/off fraction.	Crediting every fixture in the room even though only part of the zone receives adequate daylight.
Occupied daylight hours, H	Use operating schedule intersected with annual daylight availability, preferably from simulation.	Using 8,760 hours or assuming every daytime hour is sunny.
Displacement fraction, F	Derive from hourly daylight simulation plus the control sequence; confirm with trend data after occupancy.	Assuming 100% light-off whenever the sun is above the horizon.
Cooling benefit	Reduced lighting heat lowers cooling energy; calculate in the energy model. A rough screen may use 5%-20% of lighting kWh.	Counting the full lighting kWh twice.
Heating penalty	In heating climates, removed lighting heat may slightly increase heating energy.	Ignoring winter interaction when heating is electric.
Demand savings	Apply only if dimming coincides with the billing demand peak and the tariff charges for kW.	Multiplying by demand charges without verifying peak coincidence.
Maintenance	Value avoided lamp/driver replacement, lift rental, disruption, and cleaning only when truly deferred.	Claiming LED replacements that the owner would not have made within the analysis period.

Control requirement

Use automatic continuous dimming where possible. At minimum, use multilevel switching with a calibrated photosensor. Place sensors where they represent the controlled work plane but do not see the diffuser directly. Commission at different sky conditions and trend light output for several weeks.

DOE notes that properly integrated daylighting can reduce lighting energy, peak electric demand, cooling energy, and maintenance cost; it also warns that glare and excessive solar gain must be controlled. [6] These benefits are project-specific and should not be assumed as fixed percentages.

5. Economic effectiveness: cost stack and financial method

Cost component	Include in budget	Notes
Device and optics	Dome/flashing, tube kit, elbows/extensions, diffuser, suspension, security/fire accessories.	Current related rigid-kit retail examples were about $457 for 14-inch and $835 for 22-inch; commercial project assemblies differ. [8]
Roof/envelope work	Curb, deck/framing modification, membrane/flashing, insulation/energy kit, air/vapor sealing, weather protection.	Roof type, warranty requirements, access, and reroof timing can dominate cost.
Electrical controls	Photosensor, controller, dimming drivers/relays, wiring, BAS integration, programming, commissioning.	This cost is essential because controls produce the utility savings.
Soft costs	Design, structural review, permit, lift/crane, fall protection, shutdown, testing, closeout.	Commercial retrofit conditions vary widely.
Lifecycle	Cleaning/inspection, resealing if required, control calibration, diffuser replacement, roof work.	Rigid tunnels are marketed as maintenance-free; the roof interface and controls still require inspection.

Published price context, not a quote: current online listings for related VELUX rigid systems show equipment-only prices in the hundreds of dollars, while one commercial roofing cost guide gives a broad installed range of $1,500-$4,500 per tubular system. [8]-[9] A TCC/TGC project can fall outside that range because of roof curb, membrane, structure, access, fire/safety, controls, and local labor.

Recommended financial metrics

Metric	Equation / interpretation
Simple payback	Initial installed cost / first-year annual cash savings. Easy to understand, but ignores project life, escalation, discount rate, and replacements.
20-year NPV	-C0 + sum of annual net savings discounted to present value. Positive NPV means the modeled financial return exceeds the chosen discount rate.
Savings-to-investment ratio	Present value of savings / present value of costs. A ratio above 1.0 is economically favorable under the assumptions.
Break-even installed cost	Present value of expected savings. Use this as the maximum allowable installed cost during bidding.

Economic sensitivity for a $2,500 installed system

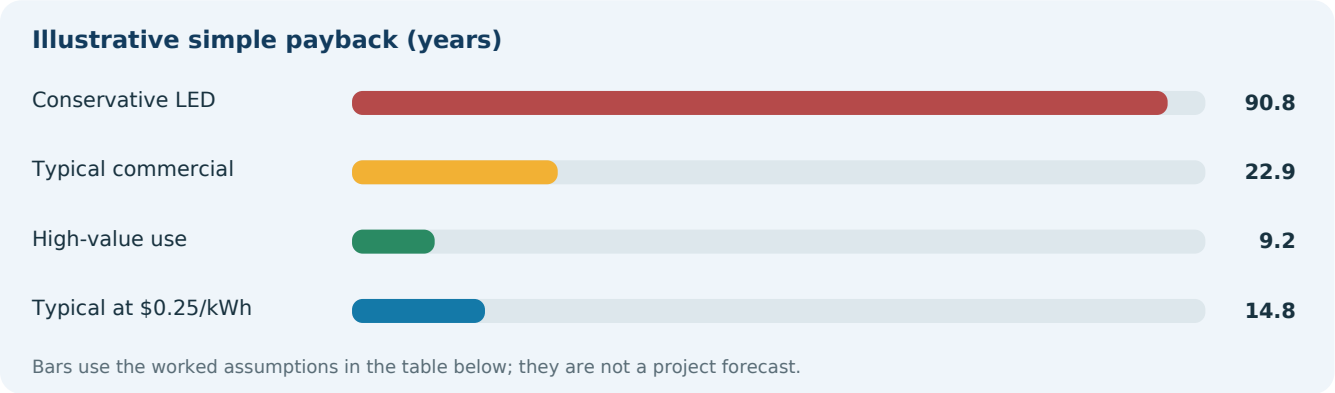
First-year annual cash savings	Simple payback on $2,500	20-year PV of savings*	20-year NPV*
$50	50.0 yr	$733	-$1,767
$100	25.0 yr	$1,467	-$1,033
$175	14.3 yr	$2,566	+$66
$250	10.0 yr	$3,666	+$1,166
$350	7.1 yr	$5,133	+$2,633

*Assumes 20 years, 5% discount rate, and 2% annual savings growth; PV factor = 14.665. Excludes residual value and replacement costs.

6. Worked economic scenarios

Illustrative U.S. commercial screening assumptions - not a site-specific prediction.

Input / result	Conservative LED	Typical commercial	High-value use
Controlled electric-light power	0.12 kW	0.30 kW	0.60 kW
Occupied daylight hours	2,200 h/yr	2,600 h/yr	3,000 h/yr
Average displacement fraction	45%	60%	70%
Lighting electricity avoided	119 kWh/yr	468 kWh/yr	1,260 kWh/yr
Cooling allowance (screening)	+10%	+10%	+10%
Net electric savings	131 kWh/yr	515 kWh/yr	1,386 kWh/yr
Energy value at $0.1341/kWh [7]	$18/yr	$69/yr	$186/yr
Maintenance + demand value	$10/yr	$40/yr	$85/yr
Total first-year cash savings	$28/yr	$109/yr	$271/yr
Installed cost used	$2,500	$2,500	$2,500
Simple payback	90.8 yr	22.9 yr	9.2 yr
20-year NPV (5%, 2% growth)	-$2,096	-$901	+$1,472



Interpretation. The device is weak as an energy-only retrofit when it displaces only one small LED fixture. It becomes economically plausible where one 22-inch unit allows a larger zone to dim for long daytime hours, where the tariff includes demand charges, or where roof work is already planned. In the typical scenario, raising electricity price from $0.1341 to $0.25/kWh reduces simple payback from 22.9 to about 14.8 years, assuming the same $40/yr non-energy cash value.

Research context. A 2025 study of a smart daylighting system using a TDD reported an operational payback around 18-20 months under its regional tariff and measured conditions. [11] That result demonstrates potential, but it should not be transferred to a U.S. VELUX project without matching system cost, tariff, climate, controls, and displaced load.

7. Implementation strategy and effectiveness checklist

Gate	Required deliverable	Proceed only if...
1. Survey	Roof plan, structure, membrane, obstructions, ceiling type, vertical distance, fire/plenum conditions, hours, tariff, lighting wattage.	A short, code-compliant route and controllable daylight zone exist.
2. Daylight model	IES/Radiance/VELUX estimate with climate file, exact tube length/elbows, room reflectance, work-plane target, and annual schedule.	Useful daylight is adequate without unacceptable glare or uniformity problems.
3. Energy model	Hourly control simulation showing fixture dimming, cooling benefit, heating penalty, and peak-demand coincidence.	Savings are traceable to the real control sequence.
4. Envelope design	Stamped details as required: curb/self-flashing, structural opening, membrane tie-in, insulation, vapor/air seal, drainage, fall protection.	Roofer, architect/engineer, manufacturer, and authority having jurisdiction accept details.
5. Financial model	Installed quote, incentives, maintenance, tariff, NPV, and sensitivity to cloudy years and control performance.	NPV/SIR meets the owner criterion and non-energy benefits are explicitly valued.
6. Commissioning	Illuminance mapping, sensor calibration, trend logs, occupant training, leak inspection, and seasonal verification.	Lights dim/off as predicted and work-plane illuminance remains acceptable.

Best-fit applications

- Warehouses, schools, retail, corridors, restrooms, and open-plan interiors occupied during daytime.
- Deep-plan areas far from facade windows.
- New construction or reroofing, where roof labor and membrane work are already mobilized.
- Spaces with high lighting wattage, long daytime schedules, high electricity/demand rates, or costly lamp access.

Poor-fit / caution

- Night-dominant occupancy or intermittently used rooms.
- Already efficient, low-wattage LEDs with no dimming interface.
- Long, offset, shaded tube route or roof crowded with equipment.
- High-humidity or pressure-sensitive buildings without a detailed air/vapor/condensation strategy.
- Roof penetrations that jeopardize structural or membrane warranties.

Measurement plan

Record pre-install lighting kW and schedules. After commissioning, trend photosensor signal, dimming command, lighting kW, and occupied hours for at least four representative weeks in different seasons. Compare measured savings with the weather-normalized model and re-commission if the shortfall exceeds the owner threshold.

8. Conclusions, due diligence, and sources

Bottom line. The Sun Tunnel/TDD is a mature passive optical daylighting device. Its technical effectiveness is strongest when the roof collector has clear sky access, the reflective path is short and straight, and the diffuser/spacing is modeled for the real work plane. Its economic effectiveness is strongest when automatic controls reduce a large electric-light load for many occupied daylight hours and the roof installation is simple or coordinated with reroofing.

Before purchasing, obtain	Why it matters
Model-specific VELUX submittal, current installation manual, NFRC/approval data, and warranty	The downloaded product sheets include older dates; current project documents govern.
Roof plan and structural opening detail signed as required	A TDD is a building-envelope penetration and may affect structure, drainage, and fire classification.
IES-based daylight simulation and control-zone drawing	Confirms quantity, spacing, work-plane illuminance, and which fixtures can be dimmed.
Line-item installed quotation	Separates device, curb/flashing, membrane, access, controls, electrical work, and commissioning.
Tariff-based lifecycle-cost calculation	Avoids relying on national average prices or generic percentage savings.

Sources and references

[1] VELUX Commercial, "Tubular Skylights." Product overview, >98% reflectivity context and 14/22-inch availability. Accessed 13 Aug 2026.

[2] VELUX Commercial, "Commercial Sun Tunnel." Applications, SunCurve, diffuser types, Flexi-Loc, and 20-year reflective warranty.

[3] VELUX Commercial, "Daylight Estimator for Sun Tunnel Skylights." IES-based geometry, reflectance, illuminance, and quantity guidance.

[4] VELUX Commercial, "Sizing Options for Commercial Sun Tunnels." TCC/TGC 14-inch and 22-inch model configurations.

[5] VELUX DASI, Commercial Sun Tunnel resource folder. Files used: TCC Product Data Sheet 060717; TGC-TMC Product Data Sheet 060717; Commercial Sun Tunnel Sizing Chart V5116-1022; current TCC/TGC installation documents.

[6] U.S. Department of Energy, "ZEB Technologies: Lighting & Daylighting." Daylight quality, 300-lux/50%-hours concept target, energy and cooling interactions.

[7] U.S. Energy Information Administration, "Prices and factors affecting prices." 2025 U.S. average commercial electricity price: 13.41 cents/kWh.

[8] Current online price examples for related VELUX rigid systems: 14-inch TGR 014 ($457) and 22-inch TGR 022 ($835). Prices are volatile and are not TCC/TGC project quotes.

[9] Chris Battaini Roofing, "Skylight Installation Commercial." Broad published tubular-system installed planning range of $1,500-$4,500; the source states figures are internet averages, not its quotations.

[10] National Fenestration Rating Council, 2026 public-review notice listing NFRC 203-2026: Procedure for Determining Visible Transmittance of Tubular Daylighting Devices.

[11] "Environmental Impact Assessment of Smart Daylighting Systems Using LCA and Measured Illuminance," Sustainability 17(18), 8463 (2025). Regional smart-TDD case; not a universal payback claim.

Professional-use limitation

Final selection and construction must be completed or reviewed by the project architect/engineer, roofing manufacturer/contractor, electrical/lighting designer, energy modeler, and local authority having jurisdiction. Product availability, codes, prices, tariffs, and warranties can change.